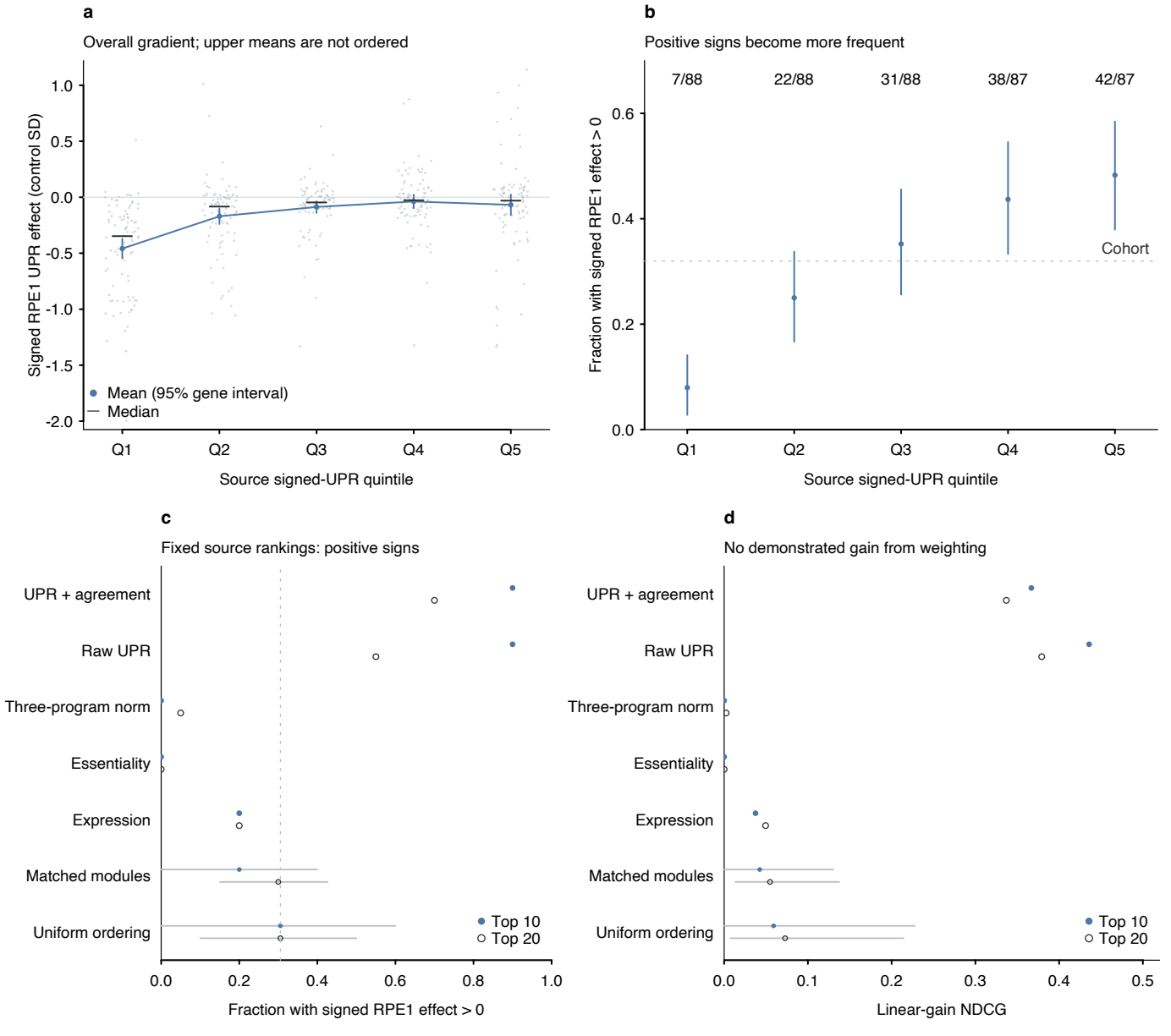

a
Overall gradient; upper means are not ordered
Signed RPE1 UPR effect (control SD)
1.0
0.5
0.0
-0.5
-1.0
-1.5
-2.0
Mean (95% gene interval)
Median
Q1
Q2
Q3
Q4
Q5
Source signed-UPR quintile
b
Positive signs become more frequent
7/88
22/88
31/88
38/87
42/87
Fraction with signed RPE1 effect > 0
0.6
0.4
0.2
0.0
Cohort
Q1
Q2
Q3
Q4
Q5
Source signed-UPR quintile
c
Fixed source rankings: positive signs
UPR + agreement
Raw UPR
Three-program norm
Essentiality
Expression
Matched modules
Uniform ordering
Top 10
Top 20
0.0
0.2
0.4
0.6
0.8
1.0
Fraction with signed RPE1 effect > 0
d
No demonstrated gain from weighting
UPR + agreement
Raw UPR
Three-program norm
Essentiality
Expression
Matched modules
Uniform ordering
Top 10
Top 20
0.0
0.1
0.2
0.3
0.4
0.5
Linear-gain NDCG